

TECHNICAL UNIVERSITY OF DORTMUND

(Technische Universität Dortmund)

Master's Program Application in Data Science

Report of the Data Analysis Project

TITLE

Data-Driven Analysis of Customer Leads and Conversion Patterns in the Real Estate Sector

Submitted by

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Declaration

I hereby declare that this report has been prepared independently and that all sources and tools used have been appropriately acknowledged.

1. Introduction

The increasing availability of data in the real estate sector enables organizations to optimize decision-making processes through analytical approaches. This project presents a data-driven analysis of customer leads to evaluate conversion behavior, customer preferences, and business performance. The primary objective is to extract actionable insights that support strategic planning, improve lead management, and enhance return on investment (ROI).

2. Dataset Description

The dataset consists of over 10,000 customer lead records obtained from a real estate company. It includes structured information related to customer interactions, property preferences, and sales outcomes.

Key attributes include:

- **Lead sources**
 - Property listing platforms (e.g., Bayut, Dubizzle, Property Finder, Meta Campaigns, Google Ads, TikTok Ads, Google Profile, Direct/Walk-in customers, Referrals)
- **Property type**
 - Commercial, Residential, Studio, Apartment, Villa, Off-plan, Secondary
- **Lead status**
 - Converted, Dropped, Not Answered, Callback, Not Interested
- **Assigned agent**
- **Temporal variables** (lead creation and follow-up timing)

3. Methodology

3.1 Data Preprocessing

Data preprocessing was performed using Python. This included handling missing values, removing duplicates, and encoding categorical variables to enable quantitative analysis.

3.2 Exploratory Data Analysis (EDA)

Descriptive statistics and visualization techniques were applied to explore patterns and distributions within the dataset.

3.3 Statistical Analysis

Correlation analysis was conducted to examine relationships between lead attributes and conversion outcomes. Additionally, a conceptual regression framework was considered.

3.4 Data Visualization

Interactive dashboards were developed using Power BI to support interpretation.

3.5 Model Evaluation and Interpretation

The Logistic Regression model was improved by addressing class imbalance using weighted classes. This resulted in a more realistic performance compared to the initial model.

The model achieved an accuracy of approximately **72.6%**. While this is lower than the initial result, it reflects a more balanced evaluation of both converted and non-converted leads.

The confusion matrix shows that the model is able to identify some converted leads; however, performance remains limited due to the highly imbalanced dataset. Specifically, only a small number of converted cases were correctly predicted, while several were misclassified. This indicates that although the model has improved, it still struggles to generalize due to insufficient representation of the minority class.



Figure 1: Model Evolution & Interpretation / Python Code

4. Power BI Reports

4.1 Lead Source Performance

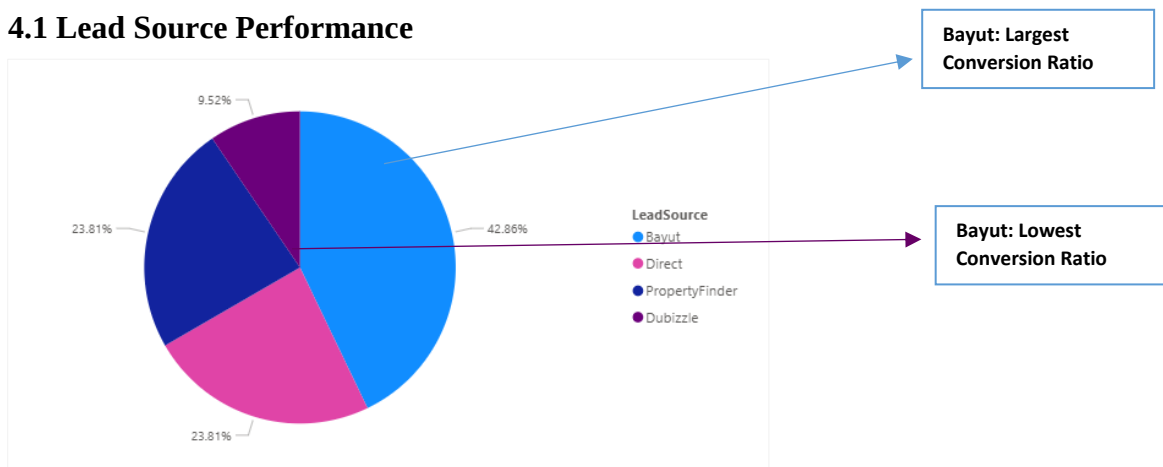


Figure 2: Lead Conversion Rate by Source

4.2 Property Type Demand

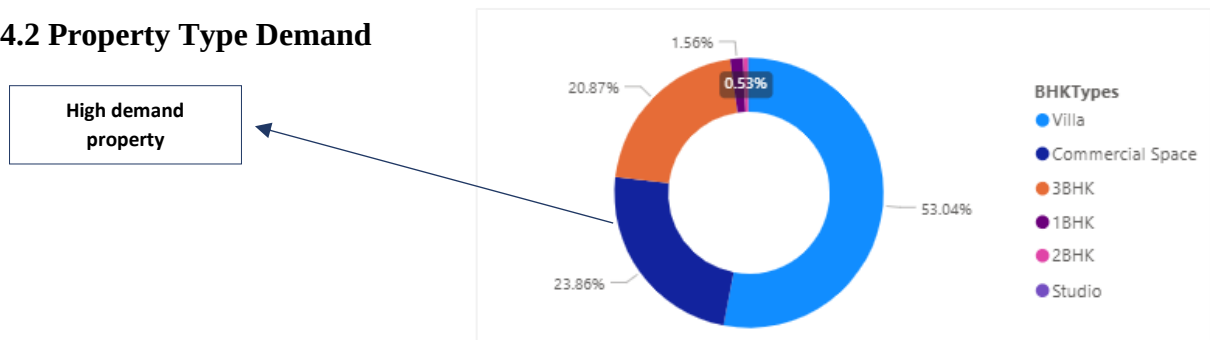


Figure 3: Distribution of Leads by Property Type

4.3 ROI Analysis

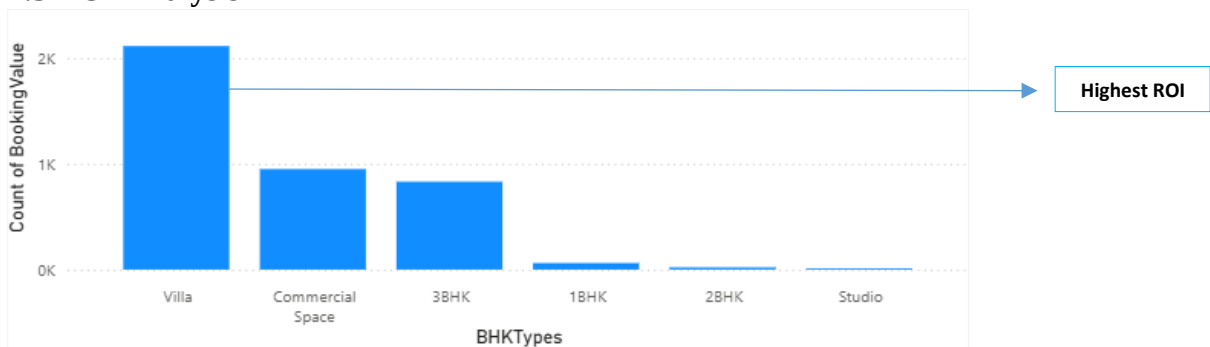


Figure 4: ROI Comparison across Property Types

4.4 Lead Drop Analysis

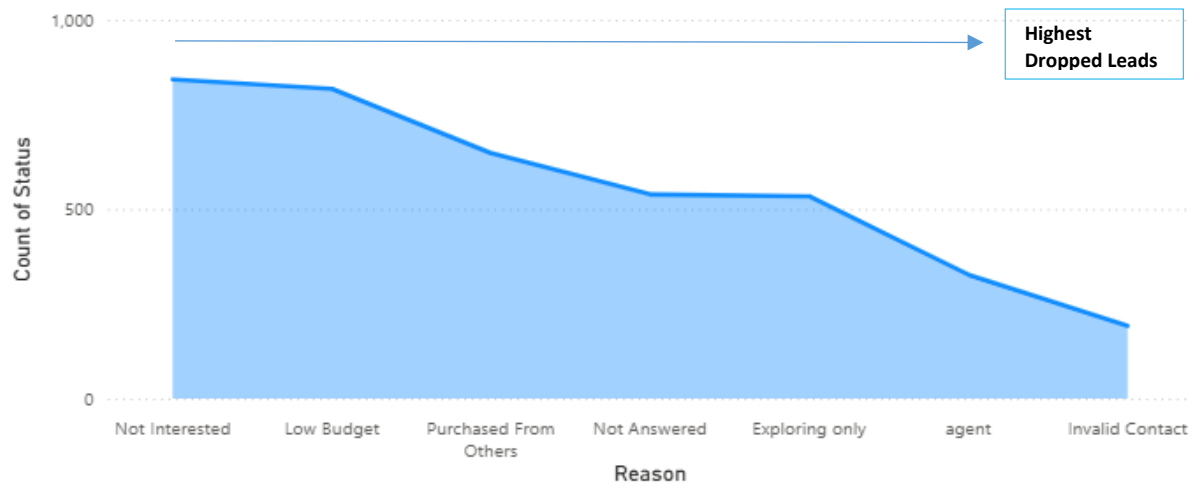


Figure 5: Reasons for Lead Drop

4.5 Microsoft Power BI Dashboard

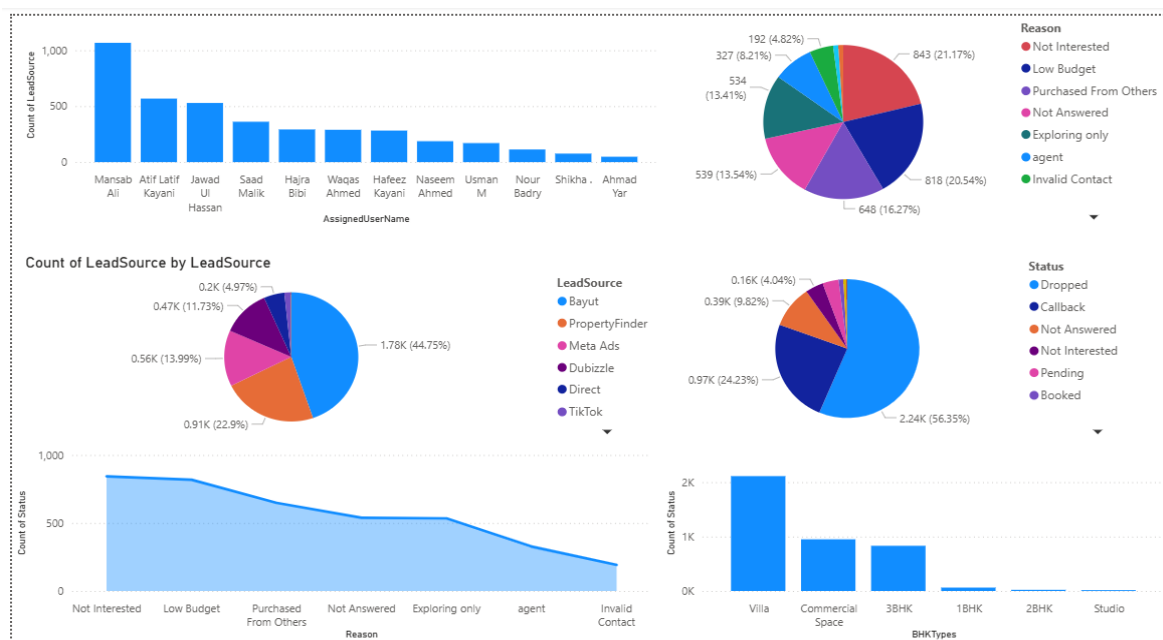


Figure 6: Power BI Dashboard

5. Discussion

The integration of statistical analysis and visualization enhances interpretability. The observed correlations and trends highlight key drivers of lead conversion and business performance.

6. Conclusion

This project demonstrates the application of data analysis techniques to real-world business data. The findings indicate that lead source and property type significantly influence conversion rates, with digital advertising channels contributing a high volume of leads but varying conversion efficiency. These insights support improved decision-making and strategic planning. This project demonstrates the application of data analysis techniques to real-world business data. The findings support improved decision-making and strategic planning.

7. Tools and Technologies

- Python
 - Microsoft Excel
 - Power BI
 - Google Colab
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8. Appendix: Sample Python Code

The following code snippet illustrates key steps in data preprocessing and basic analysis:

```
# Show first rows
print("Preview data:")
print(df.head())

# Clean data
df = df.dropna()
df = df.drop_duplicates()

# Create target column
df['converted'] = df['Status'].apply(lambda x: 1 if str(x).lower() in ['converted', 'booked'] else 0)

# Select features
X = df[['LeadSource', 'PropertyType']]
y = df['converted']

# Convert text to numbers
X = pd.get_dummies(X)

# Split data
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Train model
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000, class_weight='balanced')
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

Figure 7: Python Code

9. Key findings include

- Certain lead sources generate higher conversion rates despite lower volume
- Property type significantly influences customer engagement
- Lead drop is strongly associated with delayed follow-up

10. Limitations and Future Work

Despite the insights obtained, several limitations must be acknowledged. Firstly, the dataset is highly imbalanced, with a significantly smaller proportion of converted leads compared to non-converted ones. This imbalance affects model learning and reduces the ability to accurately predict minority class outcomes.

Secondly, the analysis is limited by the scope of available features. Important behavioral and temporal factors such as customer interaction history, response time, and agent performance were not fully incorporated. These variables could significantly improve predictive accuracy.

Additionally, the study relies primarily on structured tabular data, without incorporating advanced temporal or sequence-based patterns that may exist in customer behavior.

Future work could address these limitations by:

- Collecting a more balanced dataset with higher representation of converted leads
- Incorporating additional behavioral and time-series features
- Exploring more advanced machine learning techniques such as gradient boosting methods
- Developing a real-time prediction system for lead scoring

These improvements would enhance model performance and increase the practical applicability of the system in real-world business environments.